

$$\begin{array}{ccccccc}
 \boxed{\mathbf{H}^{(l)}} & = & \boxed{\mathbf{1}\theta_0^{(l)\top}} & + & \boxed{\mathbf{H}^{(l-1)}} & \cdot & \boxed{\left(\boldsymbol{\Theta}^{(l)}\right)^\top} \\
 N \times k_l & & N \times k_l & & N \times k_{l-1} & & k_{l-1} \times k_l
 \end{array}$$